



CAS Predictive Modeling Case Study

Renter's Insurance Pricing Strategy

Jade Liang • Dina Kruglyak • Sonia Bhatia • Thuo Noble

Table of Contents

01

**Executive
Summary**

02

**Data
Preparation**

03

**Exploratory
and Predictive
Analysis**

04

**Pure Premium
Computation**

05

**Gross Premium
Determination**

06

**Additional
Considerations**

Executive Summary

- **Goal:** Review current pricing strategy for the student tenant insurance product line
- **What We Did:**
 - Analyze demographic and claims data for 10,000 policyholders
 - Reviewed data for consistency and valid information
 - Identified opportunities to improve risk differentiation across pricing tiers
 - Proposed a new tiering mechanism based on highly relevant variables
- Since historic expense data was lacking, we researched industry average expenses and pricing was modeled based on publicly available company filings and IEE
- Generated proposals for further pricing mechanism improvements

Data Preparation

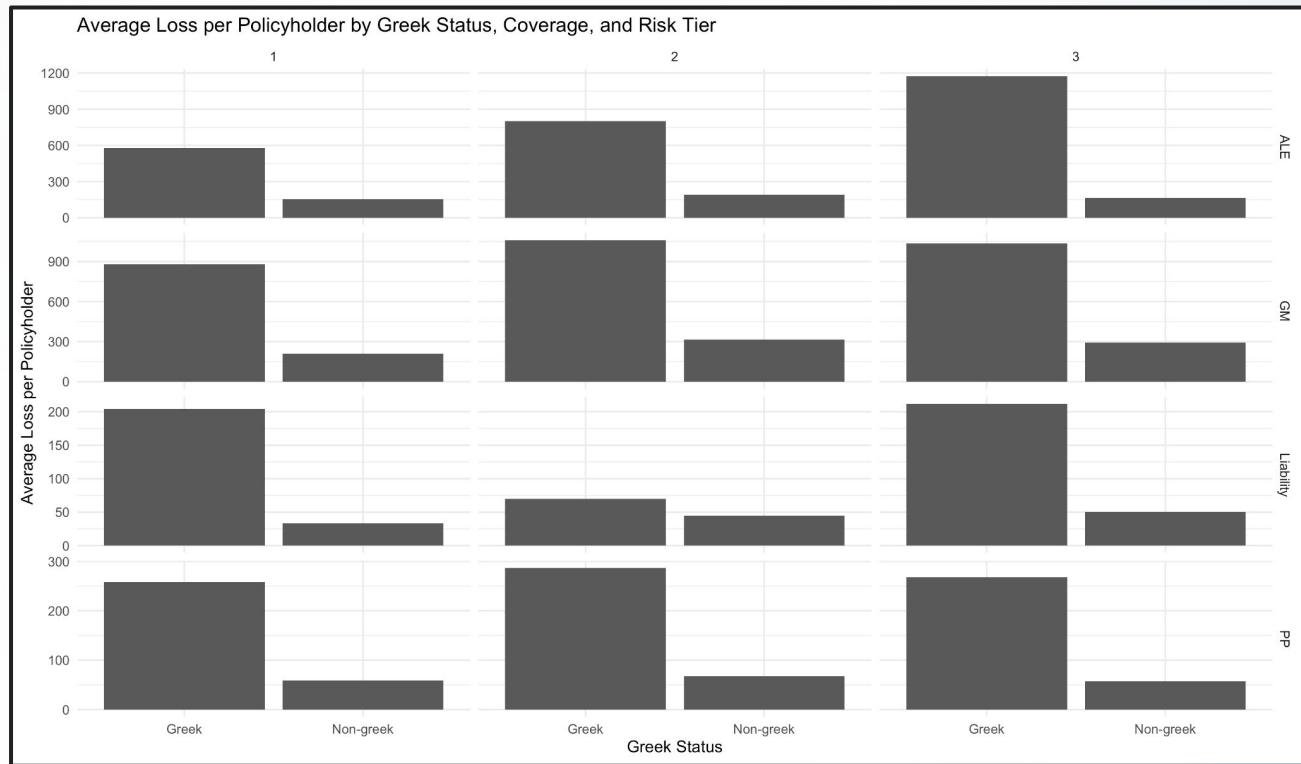
- Split data into training and holdout sets before analysis
- Checked for missing values, duplicate rows, and duplicate claim IDs
- Verified apparent repeated records were legitimate multiple claims/coverage records, not true duplicates
- Aggregated multiple claims to the policyholder-coverage level for analysis
- Reviewed structural relationships between variables
 - Class/year strongly overlaps with risk tier
 - Gender showed inconsistent loss patterns and was dropped from modeling

Exploratory Data Analysis

- **Analysis of current risk tiers:**
 - Current pricing structure successfully identifies lower-risk policyholders but presents an opportunity to improve differentiation among moderate- and higher-risk segments
- Exploratory univariable analysis shows strong differentiation of severity and frequency of claims based on the following variables:
 - GPA (3.5 or better, 3 or better, lower than 3)
 - (finer tiering based on GPA does not improve differentiation)
 - Greek vs non-Greek
 - On/off campus
 - (using incremental distance from campus does not improve differentiation)
 - Class/year (Freshman/Sophomore, Junior, Senior)

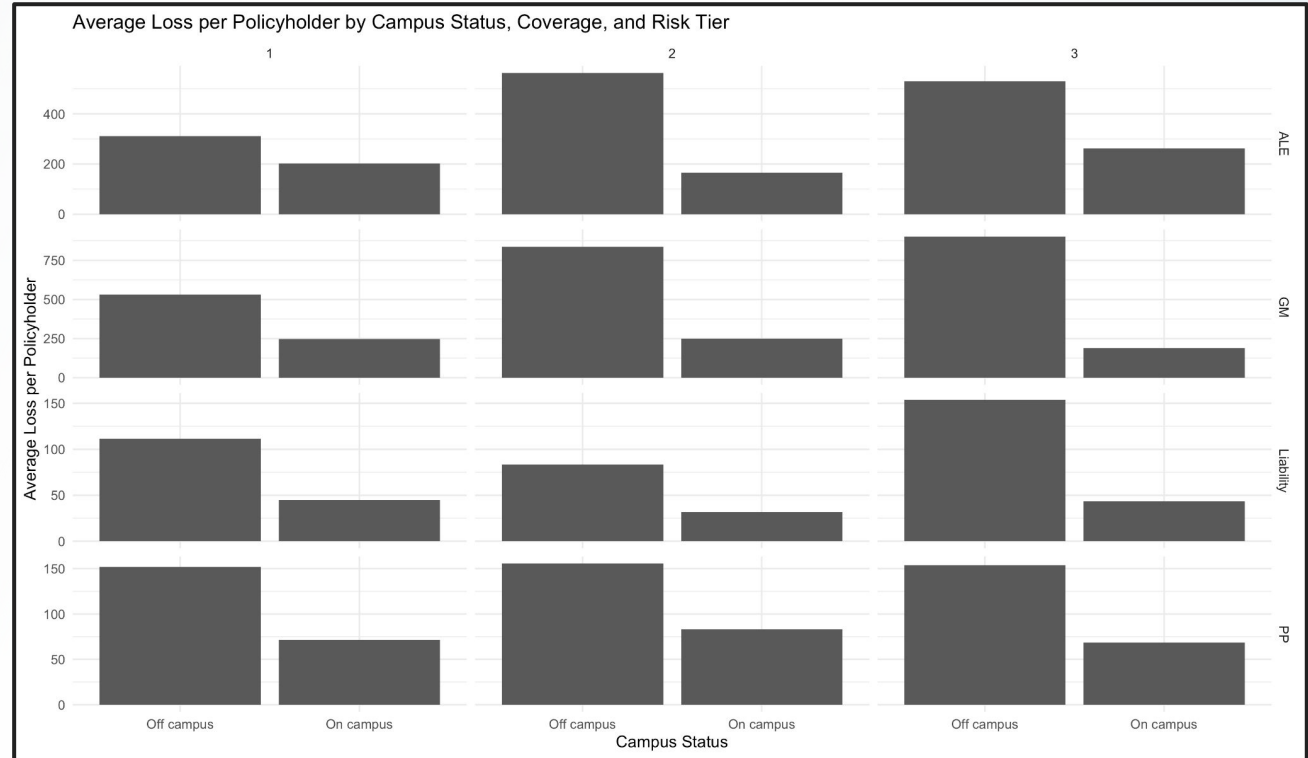
Evidence from EDA

- Greek-life policyholders showed higher average loss across coverages and tiers



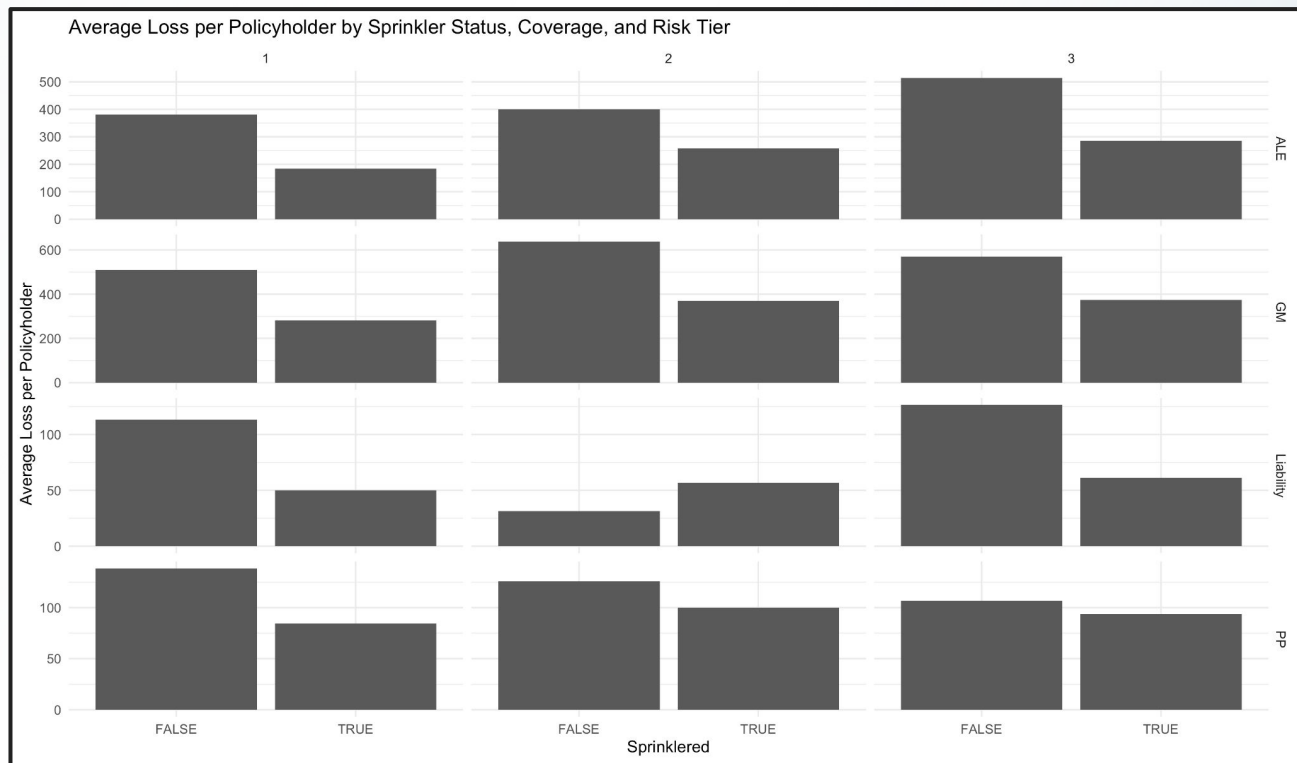
Evidence from EDA

- Off-campus policyholders generally showed higher average loss than on-campus policyholders



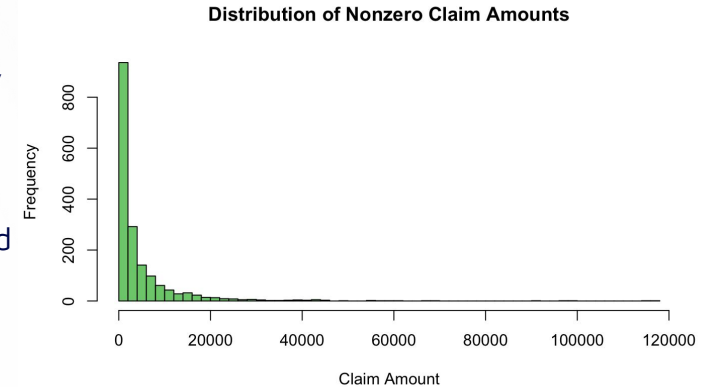
Evidence from EDA

- Sprinklered residences generally showed lower average loss



Modeling Expected Loss

- The “Amounts” column is heavily right-skewed and zero-inflated
 - Used Tweedie GLM to predict the expected claim amount per policy holder for each coverage type
- Model Assumptions:
 - Frequency and severity of claims “move in the same direction”
 - pg. 24 CAS GLM Monograph
 - The data are fully credible for every parameter
 - pg. 28 CAS GLM Monograph
 - The randomness of outcomes (i.e. amount) is uncorrelated and independent of each other
 - pg. 29 CAS GLM Monograph
- Variable Selection:
 - Used backward selection ($n = 40000 > p = 12$)
 - Performed F-test at each step
 - pg. 65 CAS GLM Monograph

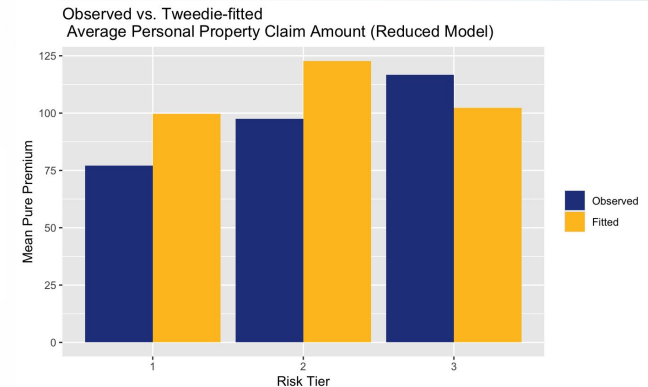
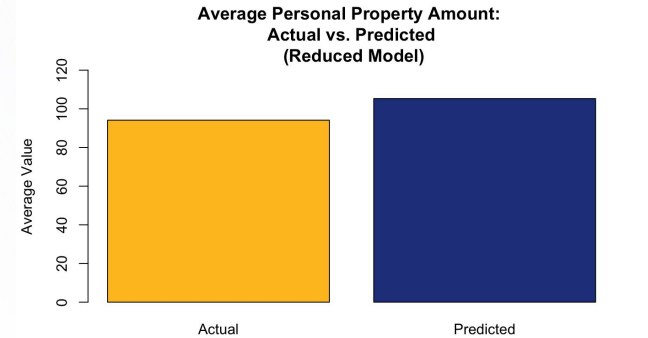


Model Results - Personal Property

Full Model			
Risk Tier	Observed	Fitted	Absolute Difference
1	\$77.15	\$99.97	\$22.82
2	\$97.44	\$123.20	\$25.76
3	\$116.77	\$102.56	\$14.22
AVG.	\$97.12	\$108.58	\$20.93

Reduced Model			
Risk Tier	Observed	Fitted	Absolute Difference
1	\$77.15	\$99.71	\$22.56
2	\$97.44	\$122.80	\$25.36
3	\$116.77	\$102.32	\$14.45
AVG.	\$97.12	\$108.28	\$20.79

```
##  
## Call:  
## glm(formula = amount ~ risk_tier + study + greek + off_campus +  
##       sprinklered, family = tweedie(var.power = p_hat, link.power = 0),  
##       data = training, control = glm.control(maxit = 15))  
""
```

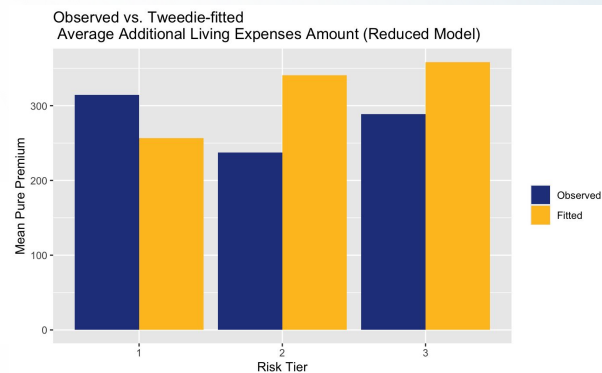
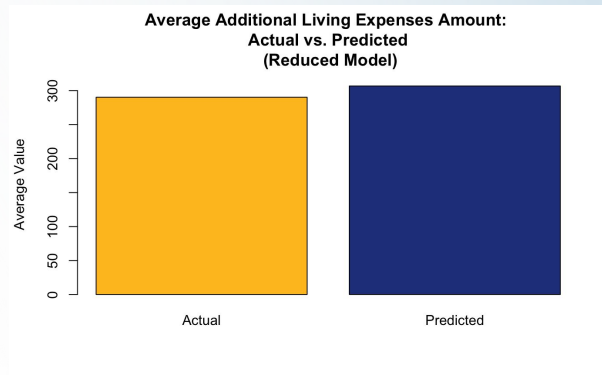


Model Results - Additional Living Expenses

Full Model			
Risk Tier	Observed	Fitted	Absolute Difference
1	\$314.63	\$255.16	\$59.48
2	\$237.51	\$342.43	\$104.92
3	\$288.54	\$357.40	\$68.87
AVG.	\$280.23	\$318.33	\$77.75

Reduced Model			
Risk Tier	Observed	Fitted	Absolute Difference
1	\$314.63	\$256.73	\$57.90
2	\$237.51	\$340.59	\$103.08
3	\$288.54	\$358.49	\$69.95
AVG.	\$280.23	\$318.60	\$76.98

```
##  
## Call:  
## glm(formula = amount ~ risk_tier + gpa + greek + off_campus +  
##       sprinklered, family = tweedie(var.power = p_hat, link.power = 0),  
##       data = training, control = glm.control(maxit = 15))
```

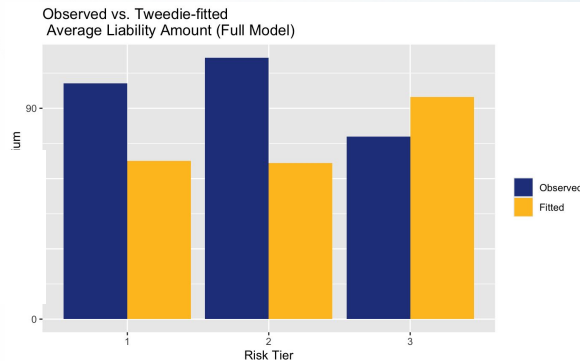


Model Results - Liability Protection

Full Model			
Risk Tier	Observed	Fitted	Absolute Difference
1	\$100.71	\$67.59	\$33.12
2	\$111.63	\$66.65	\$44.98
3	\$77.96	\$94.76	\$16.80
AVG.	\$96.77	\$76.33	\$31.63

Reduced Model			
Risk Tier	Observed	Fitted	Absolute Difference
1	\$100.71	\$67.50	\$33.21
2	\$111.63	\$66.42	\$45.21
3	\$77.96	\$94.61	\$16.65
AVG.	\$96.77	\$76.18	\$31.69

```
##
## Call:
## glm(formula = amount ~ risk_tier + study + greek + off_campus +
##       distance_to_campus + gender + sprinklered, family = tweedie(var.power = p_hat,
##       link.power = 0), data = training, control = glm.control(maxit = 15))
##
```

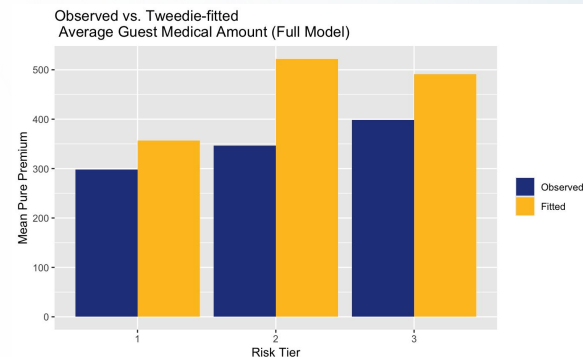
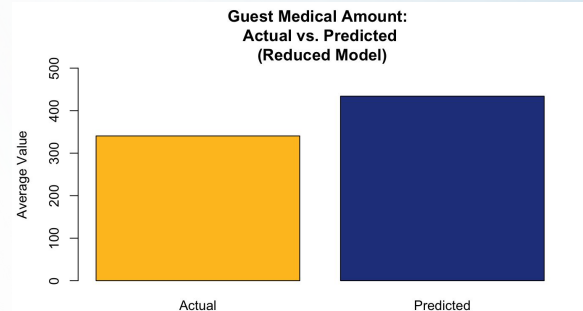


Model Results - Guest Medical Protection

Full Model			
Risk Tier	Observed	Fitted	Absolute Difference
1	\$298.29	\$360.02	\$61.73
2	\$346.72	\$532.67	\$185.95
3	\$398.04	\$491.91	\$93.87
AVG.	\$347.68	\$461.53	\$113.85

Reduced Model			
Risk Tier	Observed	Fitted	Absolute Difference
1	\$298.29	\$356.45	\$58.16
2	\$346.72	\$522.16	\$175.44
3	\$398.04	\$490.81	\$92.77
AVG.	\$347.68	\$456.47	\$108.79

```
##
## Call:
## glm(formula = amount ~ risk_tier + study + gpa + greek + off_campus +
##       distance_to_campus + sprinklered, family = tweedie(var.power = p_hat,
##       link.power = 0), data = training, control = glm.control(maxit = 15))
##
```



Gross Premium Determination

- Expense Ratio Determination
 - Industry research indicated expense ratios generally ranging from 25% to 35%.
 - A 30% Expense Ratio was chosen as a conservative base case assumption with 25% and 35% cases were tested for sensitivity
- Profit & Contingency Determination
 - A 10% assumption was selected as a reasonable midpoint for a new product with limited historical experience.
- Fixed Annual Expense Per Policy
 - Industry standards recommend a fixed expense of \$10- 20 per policy annually so a midpoint of \$15 was used as a fixed annual expense

- Sensitivity Analysis Conclusions
 - Recommended base case:
 - \$15 Fixed expense
 - 30% expense ratio
 - 10% profit & contingency provision

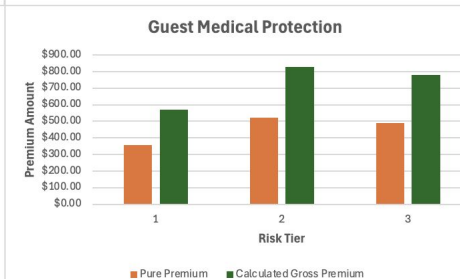
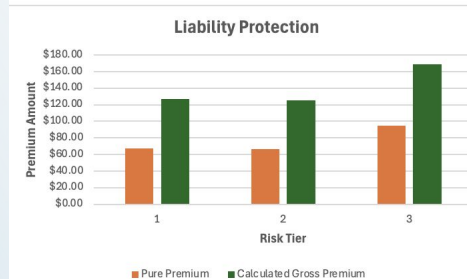
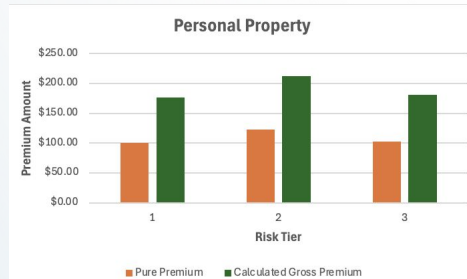
$$GP = \frac{PP + FE}{1 - E - P}$$

where:

- PP = Pure Premium
- FE = Fixed Expense Per Policy
- E = Expense Ratio
- P = Profit & Contingency Provision
- GP = Gross Premium

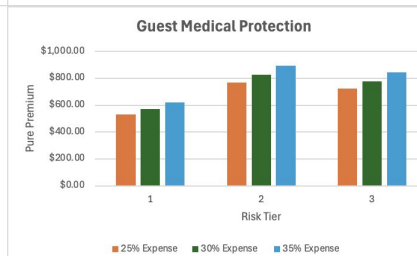
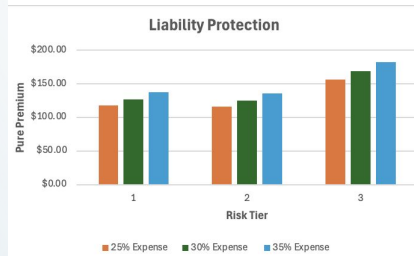
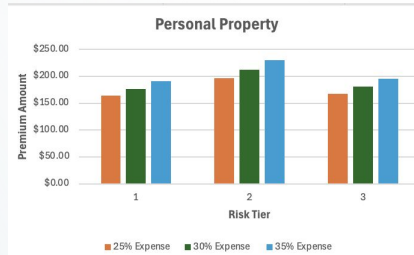
Base-Case Expense Table

Coverage	Risk Tier	Pure Premium	Fixed Expense	Variable Expense %	Profit and Contingency %	Calculated Gross Premium	Variable Expense \$	Profit and Contingency
Personal Property	1	\$99.71	\$15.00	30.00%	5.00%	\$176.48	\$52.94	\$8.82
	2	\$122.80	\$15.00	30.00%	5.00%	\$212.00	\$63.60	\$10.60
	3	\$102.32	\$15.00	30.00%	5.00%	\$180.49	\$54.15	\$9.02
Additional Living Expenses	1	\$256.73	\$15.00	30.00%	5.00%	\$418.05	\$125.41	\$20.90
	2	\$358.49	\$15.00	30.00%	5.00%	\$574.60	\$172.38	\$28.73
	3	\$318.60	\$15.00	30.00%	5.00%	\$513.23	\$153.97	\$25.66
Liability Protection	1	\$67.50	\$15.00	30.00%	5.00%	\$126.93	\$38.08	\$6.35
	2	\$66.42	\$15.00	30.00%	5.00%	\$125.26	\$37.58	\$6.26
	3	\$94.61	\$15.00	30.00%	5.00%	\$168.63	\$50.59	\$8.43
Guest Medical Protection	1	\$356.45	\$15.00	30.00%	5.00%	\$571.47	\$171.44	\$28.57
	2	\$522.16	\$15.00	30.00%	5.00%	\$826.40	\$247.92	\$41.32
	3	\$490.81	\$15.00	30.00%	5.00%	\$778.17	\$233.45	\$38.91



Sensitivity Analysis

Coverage	Risk Tier	25% Expense	30% Expense	35% Expense
Personal Property	1	\$163.88	\$176.48	\$191.19
	2	\$196.85	\$212.00	\$229.66
	3	\$167.60	\$180.49	\$195.53
Additional Living Expenses	1	\$388.19	\$418.05	\$452.88
	2	\$533.56	\$574.60	\$622.48
	3	\$476.57	\$513.23	\$556.00
Liability Protection	1	\$117.86	\$126.93	\$137.51
	2	\$116.32	\$125.26	\$135.70
	3	\$156.58	\$168.63	\$182.68
Guest Medical Protection	1	\$530.65	\$571.47	\$619.09
	2	\$767.37	\$826.40	\$895.27
	3	\$722.59	\$778.17	\$843.02



Future Strategy

- For future years, premiums would be adjusted for inflation using:

$$Premium_t = Premium_{t-1}(1 + i)$$

where:

- $Premium_t$ = Premium in year t
- $Premium_{t-1}$ = Premium in year $t - 1$
- i = Inflation Trend

- This allows us to keep up with the increases in claim costs as well as future economic changes

- Due to limited internal expense data, we would use a weighted expense approach
- Initially starting off using 10% internal expense data and 90% industry data
- With time and as the business obtains more reliable internal data we will gradually increase the % used on internal data until eventually it is relied on solely

Additional Considerations

- Model assumption of the randomness of outcomes (i.e. claim amounts) might not be true in practice
- Inconsistent data in “Gender” column
- Target audience are freshmen

Works Cited

- CAS GLM Monograph: Generalized Linear Models For Insurance Rating, Second Edition (2025 revision)
- Introduction to Predictive Modeling Using GLMs by Tevet, Dan, FCAS, MAAA (Liberty Mutual Insurance Group), and Anand Khare, FCAS, MAAA, CPCU (Milliman).
- <https://metricgate.com/blogs/tweedie-glm-insurance-pure-premium/>
- <https://www3.ambest.com/AMBV/BSRFAIFilingsWebApp/home.aspx>
- <https://www.allstatecorporation.com/resources/allstate/attachments/about/2023-4q-aic-insurance-expe...>
- https://www.casact.org/sites/default/files/old/studynotes_werner_modlin_ratemaking.pdf